

MoonBuddy: A Voice-based Augmented Reality User Interface That Supports Astronauts During Extravehicular Activities

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Figure 1: We present a voice-based AR interface for autonomous space exploration, called MoonBuddy. *Left:* A user sampling rocks using the MoonBuddy AR assistance at Johnson Space Center. *Right:* User's field of view on MoonBuddy's navigational interface, including a compass (top), collapsed task list (top left) and vitals panel (top right), and a map view (bottom right).

ABSTRACT

As NASA pursues Artemis missions to the moon and beyond, it is essential to equip astronauts with the appropriate human-autonomy enabling technology necessary for the elevated demands of lunar surface exploration and extreme terrestrial access. We present MoonBuddy, an application built for the Microsoft HoloLens 2 that utilizes Augmented Reality (AR) and voice-based interaction to assist astronauts in communication, navigation, and documentation on future lunar extravehicular activities (EVAs), with the goal of reducing cognitive load and increasing task completion. User testing results for MoonBuddy under simulated lunar conditions have been positive overall, with participants indicating that the application was easy to use and helpful in completing the required tasks.

CCS CONCEPTS

Human-centered computing → Mixed / augmented reality;
Natural language interfaces.

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KEYWORDS

Augmented Reality, voice user interface, spacesuit interface

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1 INTRODUCTION

Historically, EVAs have been highly choreographed and dependent on constant communication with the Mission Control Center (MCC) to provide updates and guidance on mission-critical information on tasks, navigation, and emergency procedures. As the backbone of space exploration missions, EVA systems must be as safe and efficient as possible to ensure that the crew meets their mission goals of exploration and acquiring samples. However, as NASA pursues long-term space exploration on Artemis missions, communication delays between astronauts and the MCC will increase significantly due to the distance between the MCC and the respective planet; messages take 1.3 seconds to reach the moon, and up to 20 minutes to reach Mars [9, 11]. These delays necessitate solutions that equip astronauts with the appropriate autonomy-enabling technology

necessary for the elevated demands of lunar surface exploration and extreme terrestrial access.

On Apollo missions, each astronaut had a paper-based cuff checklist of written EVA instructions to provide support independent of the MCC. While updates to this format have been made over the years in the form of digital wrist displays [1], diverting attention to the wrist while navigating unfamiliar landscapes and executing tasks can negatively impact a user's cognitive load, situational awareness, and overall efficiency. As an alternative, NASA has invested in AR as a solution, which is currently being tested on the International Space Station (ISS) by utilizing teleconferencing capabilities, real-time AR annotations, and 3-D holograms [2]. For future missions, NASA's NEEMO and DAVD projects focused on prototyping AR heads-up-displays (HUDs) for use during EVAs, with NASA's custom AR-integrated spacesuit helmet, called Joint Augmented Reality Visual Informatics System (JARVIS), in mind [1, 3, 6]. AR technology allows real-time information to be visually overlaid on top of astronauts' environment and assists with a variety of tasks. [6]. In contrast to AR solutions on earth, however, astronauts face unique challenges in terms of input capabilities and output requirements, including restricted movement and harsh lighting conditions caused by direct illumination from the Sun.

We present a step towards an integrated AR system for lunar EVAs, called MoonBuddy. MoonBuddy is a voice-based AR application that integrates navigation, task assistance, documentation tools, and hazard avoidance through geometry-aware surface highlighting. As the main input paradigm, the system allows the user to control UI elements with short voice commands, leveraging the benefits of voice-based interaction in AR for manipulation [10] and general communication [5]. We detail the design rationale and implementation, as well as a preliminary evaluation of MoonBuddy at Johnson Space Center in Houston, Texas, in the dedicated lunar simulation area (the "rock yard"), as part of the NASA Spacesuit User Interface Technologies for Students (SUITS) Challenge [4].

2 MOONBUDDY

The goal of MoonBuddy is to aid autonomous lunar EVAs, specifically for following tasks: navigation, scientific data collection, terrain sensing, and Lunar Search and Rescue (LSAR).

2.1 Design requirements

To create a set of design requirements, we conducted a series of semi-structured expert interviews with former astronaut Jerome Apt and multiple stakeholders from the NASA team. Additionally, we analyzed the publicly accessible audio files of former Apollo missions to discover common pain points the astronauts faced.¹ We distilled the following design requirements for MoonBuddy.

- (1) MoonBuddy must be operable while users are wearing EVA gloves and executing dexterous tasks.
- (2) MoonBuddy shall be unobtrusive, easy to use, and decrease the cognitive load of users.
- (3) Astronauts must be able to access EVA task instructions and suit vitals at any time.

- (4) MoonBuddy shall be flexible enough to handle astronaut requests at any time.
- (5) MoonBuddy should provide real-time navigation between any given points or crew members.
- (6) Astronauts must be able to communicate with EVA system crews without interference.
- (7) MoonBuddy should provide environmental awareness of hazardous regions.
- (8) MoonBuddy should clearly display warning messages for important anomalies.

2.2 System overview

Based on those requirements, we designed MoonBuddy as a voice-only user interface. A first-person screenshot of the system is shown in Figure 1.

2.2.1 Voice user interface. One of our primary concerns when developing MoonBuddy was identifying the appropriate interaction modality. While gestural interactions are a common form of AR interaction, they are arguably not appropriate given the mobility constraints of the spacesuits and interfere with tasks that use both hands, such as geological sampling. Similarly, explicit interaction through eye-tracking was rejected due to the potential of false positives. Therefore, a voice user interface (VUI) was selected to provide a hands-free solution that integrated into the voice-loop framework that astronauts are accustomed to. Specifically, astronauts can use voice commands to manipulate UI elements from navigation aids, to mission instructions and emergency responses.

2.2.2 Visual user interface. MoonBuddy displays an always-on panel that contains the list of possible voice commands and a compass. Because an astronaut's attention is typically directed toward the lunar surface, this panel is located above the users' field of view (FoV), at a fixed height, to avoid obstructing their environment. They can look up to reference the panel information when needed. Through voice commands, astronauts can open detail windows such as a map, an expanded task list, or photos and videos. Detail windows are displayed as view-anchored head-up display items and can be toggled on-demand.

Navigation. During lunar EVAs, astronauts must travel to different locations for scientific data collection. Thus, MoonBuddy provides real-time navigation for long-range (from point A to point B) and short-range targets (obstacle avoidance). Users can activate long-range navigation with a voice command, "Go to [Location Name]". Our system will then generate safe paths using data from lunar surface maps and on-site spatial mapping. Astronauts are guided to their destination via the compass and on-demand top view map. In addition to long-range navigation, MoonBuddy uses the HoloLens depth camera to provide astronauts with real-time highlighting of dangerous areas and assist vision in adverse lighting scenarios for short-range navigation and obstacle avoidance.

Science sampling. MoonBuddy guides astronauts through the process of recording observations of selected lunar samples. Key considerations within this task are to allow flexibility in choosing what types of samples to document, and to support users with varying levels of geological expertise. MoonBuddy allows astronauts to identify the sample on-site that they wish to document

¹Apollo Audio files. Last accessed July 11th 2022. <https://www.hq.nasa.gov/alsj/alsj-ApolloAudio.html>.

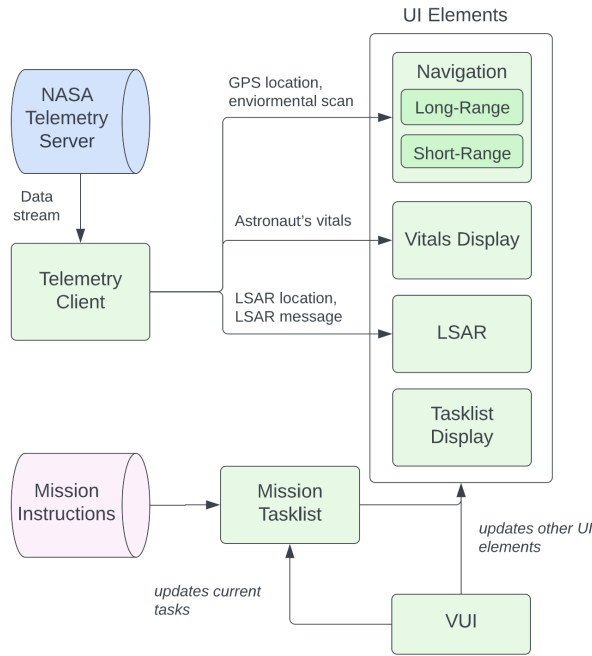


Figure 2: Information flow for the MoonBuddy system.

and subsequently leads them through a collection procedure sequence, including taking photos, in the visual tasklist interface. The instructions closely follow NASA’s sampling procedures [4] and are detailed enough for astronauts who are not trained scientists to effectively document the required information, while also allowing experts to quickly skip through it.

2.2.3 Lunar search and rescue (LSAR). MoonBuddy’s LSAR system allows astronauts to set distress beacons and alert crew members of emergency situations. Rescue messages from distress beacons show the astronaut’s condition and current location relative to their position to prompt rescue. Confirming the rescue prompt will bring up long-range navigation mode and guide the crew member toward the distressed astronaut’s location. LSAR is currently triggered manually but should alert crew members of emergencies based on user vitals and fall detection in the future.

2.3 Implementation

MoonBuddy was created using Unity3D for the Microsoft HoloLens 2. The system, illustrated in Figure 2, is divided into six major components: telemetry stream, voice interface, mission task list, navigation, vitals display, and Lunar Search and Rescue. We use the Microsoft Mixed Reality Toolkit (MRTK) for the voice interface and the following behavior of world-space UIs, as well as rendering of the interface elements. The VUI works via MRTK text recognition, where each voice command is mapped to a corresponding function that can perform actions across different systems. Additionally, we interface through HoloLens’ environmental scans through MRTK to extract the world models. As a data source, MoonBuddy interfaces with a NASA telemetry server that provides a data stream of the astronaut’s vitals, the astronaut’s GPS location, and LSAR emergency

messages. To simulate the data latency between the Earth and the Moon, data is updated every 5 seconds. Long-range navigation uses GPS coordinates from the telemetry stream and user orientation to update the map display and HUD compass. Short-range navigation uses the scanned terrain data and processes it with an edge detection shader. Vitals displays via a collapsible UI interface. Finally, LSAR is an independent system triggered by telemetry information and can override existing processes in the case of an emergency.

3 PRELIMINARY EVALUATION

We performed a preliminary evaluation at the Johnson Space Center, Houston, TX, USA, in an area that simulates the lunar surface and its lighting conditions (dubbed the "rock yard"), shown in Figure 3. The testing occurred during two sessions between 7 pm and midnight with an expert from the NASA team (one space exploration systems engineer and one principal investigator from NASA’s human interface branch; both with more than ten years of experience in aerospace engineering and research).



Figure 3: Various test areas within the rock yard.

3.1 Tasks and procedure

During testing, participants used MoonBuddy to navigate to three different geological sampling sites and collected rock samples. During the session, an LSAR message was sent through the NASA telemetry server, and participants had to react accordingly. Experimental sessions were preceded by a 45-minute training and concluded with a debriefing session. Feedback was gathered using video recordings and a think-aloud protocol.

3.2 Results

Participants appreciated the simple voice commands and the guided nature of MoonBuddy, especially during the geological sampling tasks, which fully occupied participants’ hands. As shown in Figure 3, the testing area mimicked the lighting conditions of the southern lunar hemisphere, resulting in very bright regions and hard surface shadows. Participants noted that adaptive brightness for user interface elements would be beneficial to balance visibility.

Additionally, participants positively commented on the surface edge detection and that it helped them avoid hazards during navigation. They also noted that AR navigation should be anchored in the real world, either through overlays that show the desired path

in the user's FoV, or having a realistic map of the environment. In an environment lacking landmarks, AR could become of assistance by overlaying virtual landmarks to help with navigation.

4 DISCUSSION AND CONCLUSION

With its ability to assist in user navigation, vitals monitoring, task performance, and lunar search and rescue, MoonBuddy has shown how an AR interface can be a valuable addition to space missions. Our system successfully demonstrated the viability and capability of voice-controlled AR interfaces on extraterrestrial surfaces. Preliminary testing highlighted the importance of taking lunar lighting conditions and user capabilities into account, which differ for astronauts than when designing for users on earth. The voice-based interface enabled users to interact with the system in a hands-free manner, which is of high importance for our specific context. The assistive role of the interface enabled increased autonomy, which will be a key factor for future NASA missions.

In the future, we want to further experiment with AR's ability to directly interact with the world and experiment with the impact of AR landmarks on navigation tasks in a monotonous environment, while exploring different navigation approaches [7]. We hope to make the AR interface adaptive not only to the environment users are in, but also to the mental state of the user, e.g., adapting the interface based on users' cognitive load [8]. We aim to implement more advanced geological sampling abilities, such as panoramic photos, and categorize previously collected data. We strongly believe AR has a place in the future of space exploration to support astronauts' autonomy, performance, and safety.

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REFERENCES

- [1] Steven Chappell, Kara Beaton, Carolyn Newton, Trevor Graff, K. Young, David Coan, Andrew Abercromby, and Michael Gernhardt. 2017. Integration of an Earth-based science team during human exploration of Mars. 1–11. <https://doi.org/10.1109/AERO.2017.7943727>
- [2] David Coan. 2015. *NEEMO 20 Exploration EVA Results*. Technical Report.
- [3] David Coan. 2019. Relevant Environments for Analysis and Development (READY): Enabling Human Space Exploration Through Integrated Operational Testing. In *International Conference on Environmental Systems (ICES 2019)*-11.
- [4] Trevor G. Graff. 2022. NASA SUITS 2022 Mission Briefing #4 AR for Lunar Spacewalks. https://www.youtube.com/watch?v=20RcGiXaxsY&ab_channel=NASASTEM
- [5] Ed Kaiser, Alex Olwal, David McGee, Hrvoje Benko, Andrea Corradini, Xiaoguang Li, Phil Cohen, and Steven Feiner. 2003. Mutual disambiguation of 3D multimodal interaction in augmented and virtual reality. In *Proceedings of the 5th international conference on Multimodal interfaces* (Vancouver, British Columbia, Canada) (*ICMI '03*). Association for Computing Machinery, New York, NY, USA, 12–19. <https://doi.org/10.1145/958432.958438>
- [6] Briana L Krygier. 2021. *Joint Augmented Reality Visual Informatics System Project: HII Prototype System Goals for FY22*. Technical Report.
- [7] Jaewook Lee, Fanjie Jin, Younsoo Kim, and David Lindlbauer. 2022. User Preference for Navigation Instructions in Mixed Reality. In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*. 802–811. <https://doi.org/10.1109/VR51125.2022.00102>
- [8] David Lindlbauer, Anna Maria Feit, and Otmar Hilliges. 2019. Context-Aware Online Adaptation of Mixed Reality Interfaces. In *Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology* (New Orleans, LA, USA) (*UIST '19*). Association for Computing Machinery, New York, NY, USA, 147–160. <https://doi.org/10.1145/3332165.3347945>
- [9] Stanley G. Love and Marcum L. Reagan. 2013. Delayed voice communication. *Acta Astronautica* 91 (2013), 89–95. <https://doi.org/10.1016/j.actaastro.2013.05.003>
- [10] Thammathip Piumsomboon, David Altimira, Hyungon Kim, Adrian Clark, Gun Lee, and Mark Billingham. 2014. Grasp-Shell vs gesture-speech: A comparison of direct and indirect natural interaction techniques in augmented reality. In *2014 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. 73–82. <https://doi.org/10.1109/ISMAR.2014.6948411>
- [11] Marshall Smith, Douglas Craig, Nicole Herrmann, Erin Mahoney, Jonathan Krezel, Nate McIntyre, and Kandyce Goodliff. 2020. The Artemis program: an overview of NASA's activities to return humans to the moon. In *2020 IEEE Aerospace Conference*. IEEE, 1–10.